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28/3/2026 (Day 40).

Program:

```
~~import~~
# data collection
df = pd.read_csv("student.csv")
df1 = pd.read_excel("customers.csv")
df2 = sns.load_dataset("titanic")
from sklearn.datasets import load_iris
df = load_iris()
x = iris.data
y = iris.target

# data preparation
df.head()
df.tail()
df.info()
df.describe()
df.shape
df.dtypes
df.size
df.size

# data cleaning:
df.isnull().sum()
df.duplicated().sum()

# train_test_split
from sklearn.model_selection import train_test_split
X_train, y_train, x_test, y_test = train_test_split(x, y,
                test_size = 0.2, random_state = 42)
```

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```
# regression ---> linear regression
from sklearn. ~~learn~~ linear_model import LinearRegression
model = LinearRegression().
model.fit (x_train, y_train)
y_pred = model.predict (x_test).

# ridge
from sklearn.linear_model import Ridge
model = Ridge (alpha = 1,0)
model.fit (x_train, y_train)

# polynomial
from sklearn.preprocessing import PolynomialFeatures
model = PolynomialFeatures (degree = 3).
model.fit_transform (x)

# Logistic Regression
from sklearn.linear_model import LogisticRegression
model = LogisticRegression
model.fit (x_train, y_train).

# ~~fr~~ K-nearest neighbors
from sklearn.neighbors import KneighborsClassifier
model = KNeighborsClassifier (n_neighbors = 5).
model.fit (x_train, y_train).

# support vector machine
from sklearn. ~~neighbors~~ svm import SVC
Model = SVC (Kernel = 'rbf')
model.fit (x_train, y_train).
```

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```
#descision Tree
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier()
model.fit (x_train , y_train)

#random forest classifier
from sklearn.ensemble import RandomForestClassifier
model=RandomForestClassifier()
model.fit (x_train, y_train)

#naive byes.
from sklearn.naive_byes import GaussianNB
model=GaussianNB()
model.fit (x_train , y_train)

# Kmeans.
from sklearn.cluster import KMeans.
model = KMeans (n_clusters=3)
model.fit (x)
labels=model.predict (x).

# dbscan
from sklearn.cluster import DBSCAN
model=DBSCAN (eps=0.5, min_samples=5).
labels=model.predict (x).

#Gaussian mixture model
from sklearn.mixture import GaussianMixture
model=GaussianMixture (n_components=3).
model.fit (x)
labels=model.predict (x).
```

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```
# hiearchical
from sklearn.cluster import AgglomerativeClustering
model = AgglomerativeClustering().
labels = model.fit_predict(x).
```

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